

Mehodology used for acoustic assessment of IEGJM capelin

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Presentation given on 'Advance acoustics course' at MFRI oct 2026



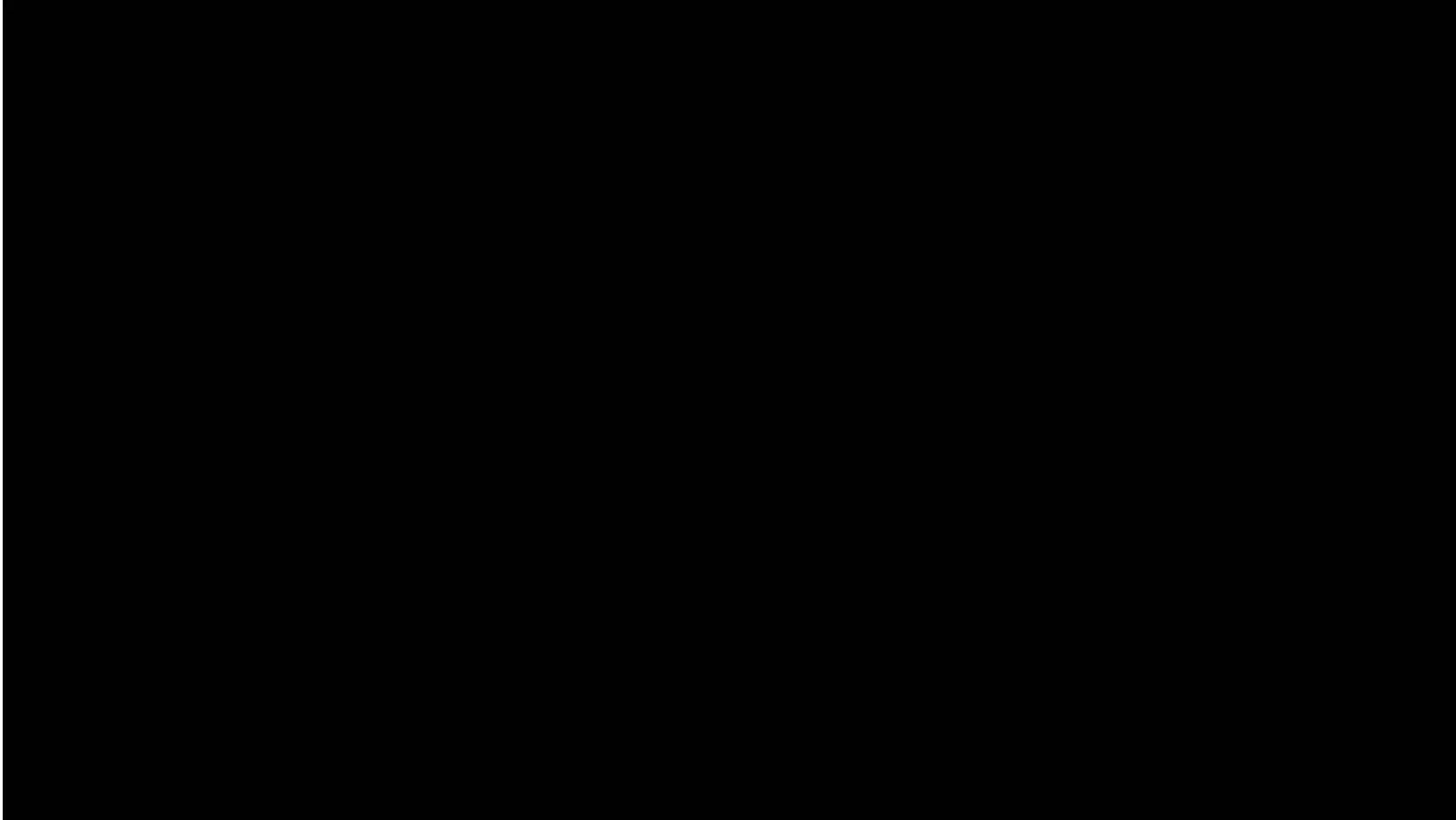
Summary

- Introduction
- Survey design
- Execution of survey
- Analysis of results
 - Bootstrap / slembivalshermanir með endurísetningu
- ‘Changes to methodology’
- Conclusions

Introduction



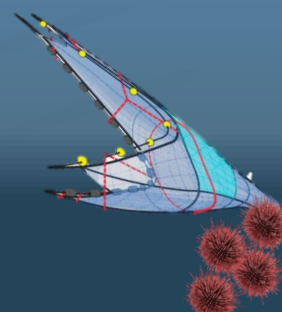
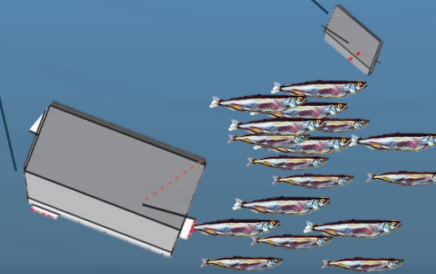
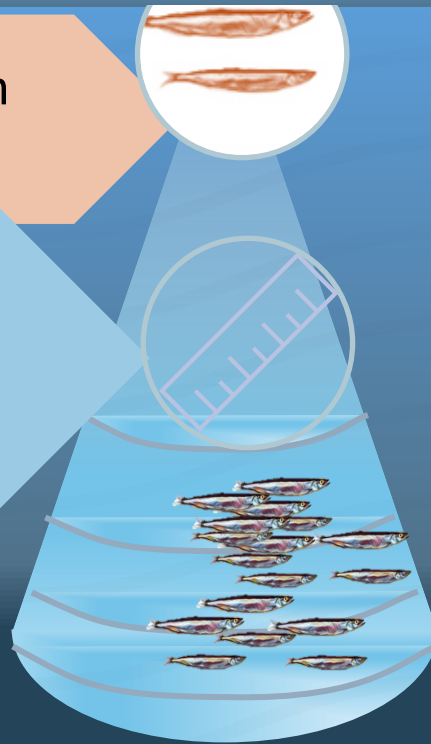
Capelin surveys – design and analysis of results





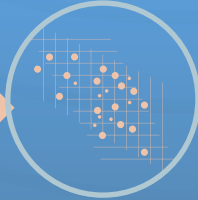
Capelin numbers and biomass are estimated based on acoustic registration and ground truthing from pelagic trawl samples

Relationship of target strength (TS) and length are key in the conversion of acoustic data to numbers of fish

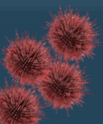
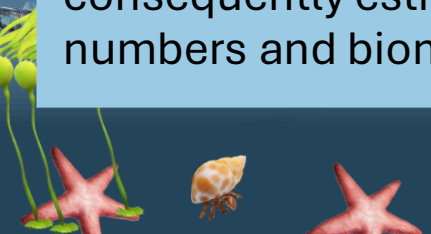
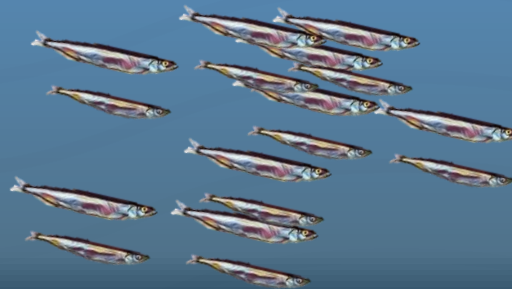




Target strength (TS) is the backscatter from a single fish



Capelin length, weight, condition, swimbladder size, tilt, maturation and fat content, influence TS and consequently estimates of numbers and biomass



Survey design



Objectives in winter 2025

were estimating the size of the maturing part of the stock (SSB) with acoustics survey with systematic design.

Survey tracks and results from 4 coverages

Jan 15-28

Jan 27-31

Feb 8-19

Feb 24-28

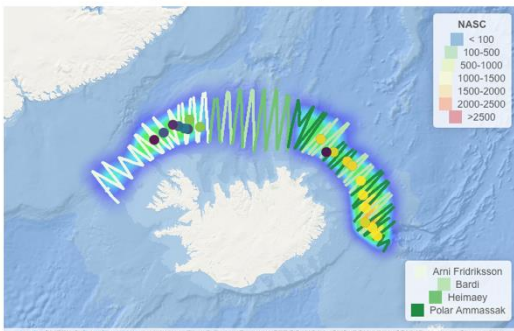


Figure 6: Capelin distribution as the relative density of acoustic backscatter as NASC and proportion mature within capelin sample during the survey in January 2025. Survey tracks are shown in different colors. Proportions below 0.5 are shown as black to green points, while proportions above 0.5 to 1 in light green to yellow, respectively.

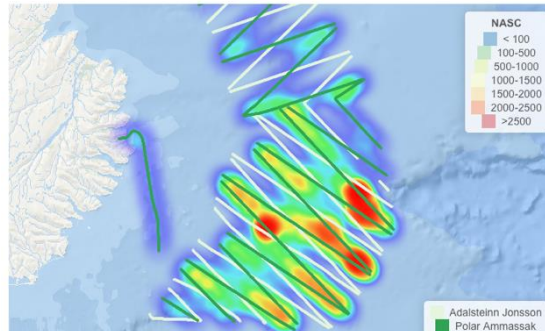


Figure 7: Capelin distribution as relative density of acoustic backscatter as NASC during the survey from 27 to 31 January 2025. Survey tracks are shown in different colors.

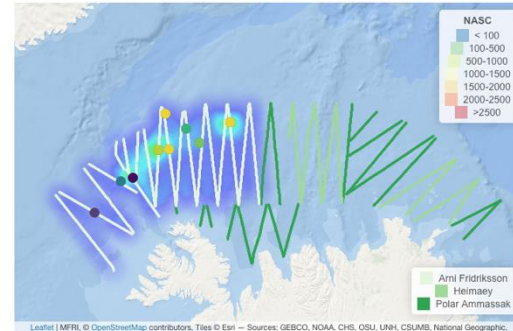


Figure 8: Capelin distribution as relative density of acoustic backscatter as NASC and proportion mature within capelin sample during the survey from 8 to 19 of February 2025. Survey tracks are shown in different colors. Proportions below 0.5 are shown as black to green points, while proportions above 0.5 to 1 in light green to yellow, respectively.

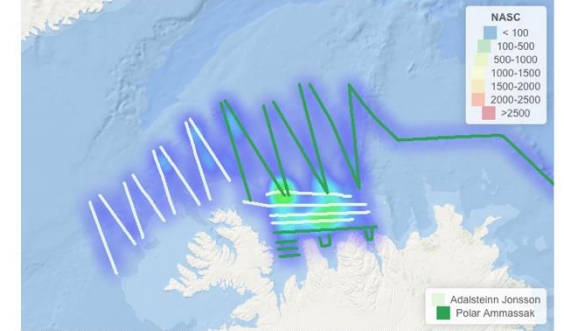
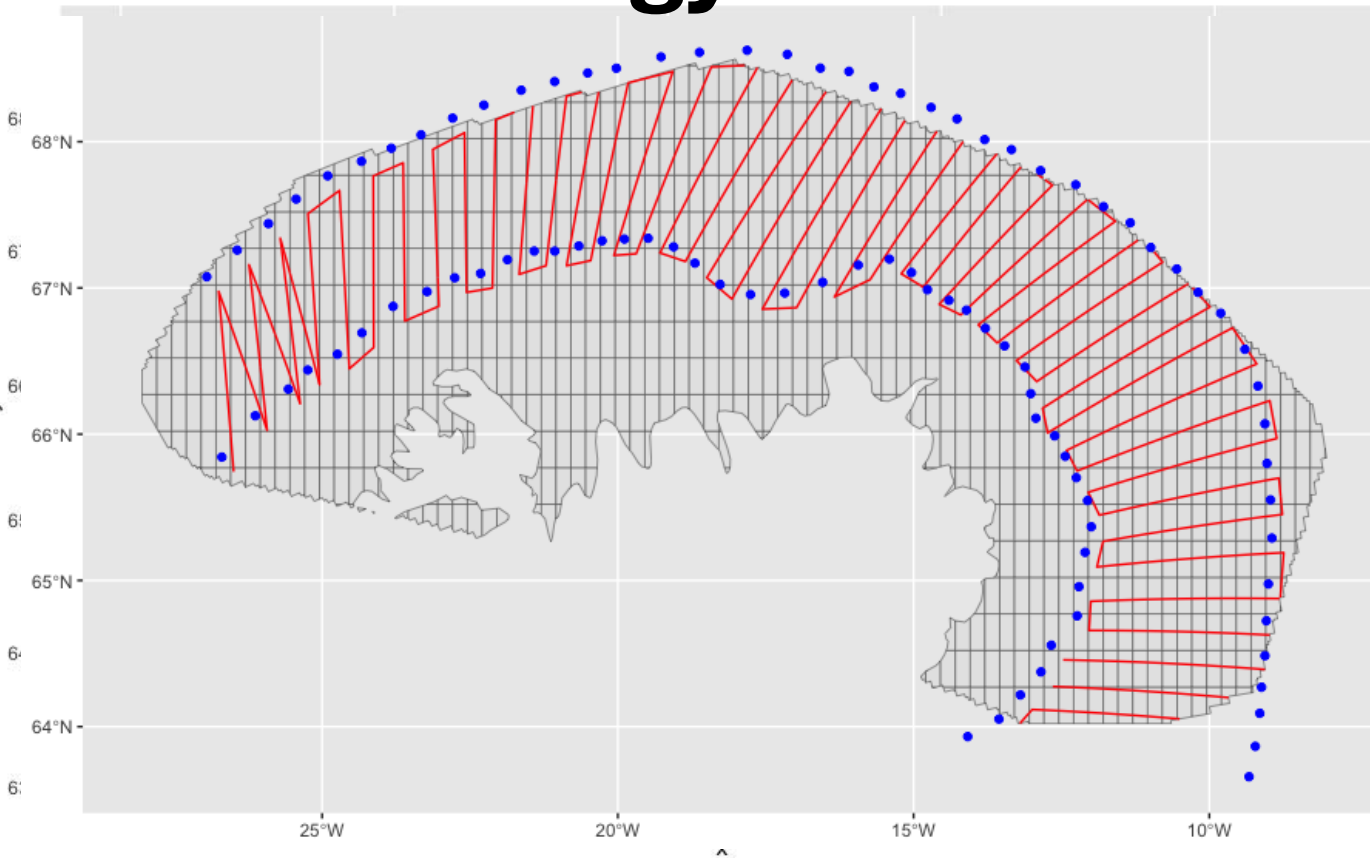


Figure 9: Capelin distribution as relative density of acoustic backscatter as NASC and proportion mature within capelin sample during the survey from 8 to 19 of February 2025. Survey tracks are shown in different colors.

Survey design:

random start, systematic track layout.

Methodology



Previous
surveys (2000-
2021)

Define
coverage
from capelin
registrations
(**nasc > 0**).

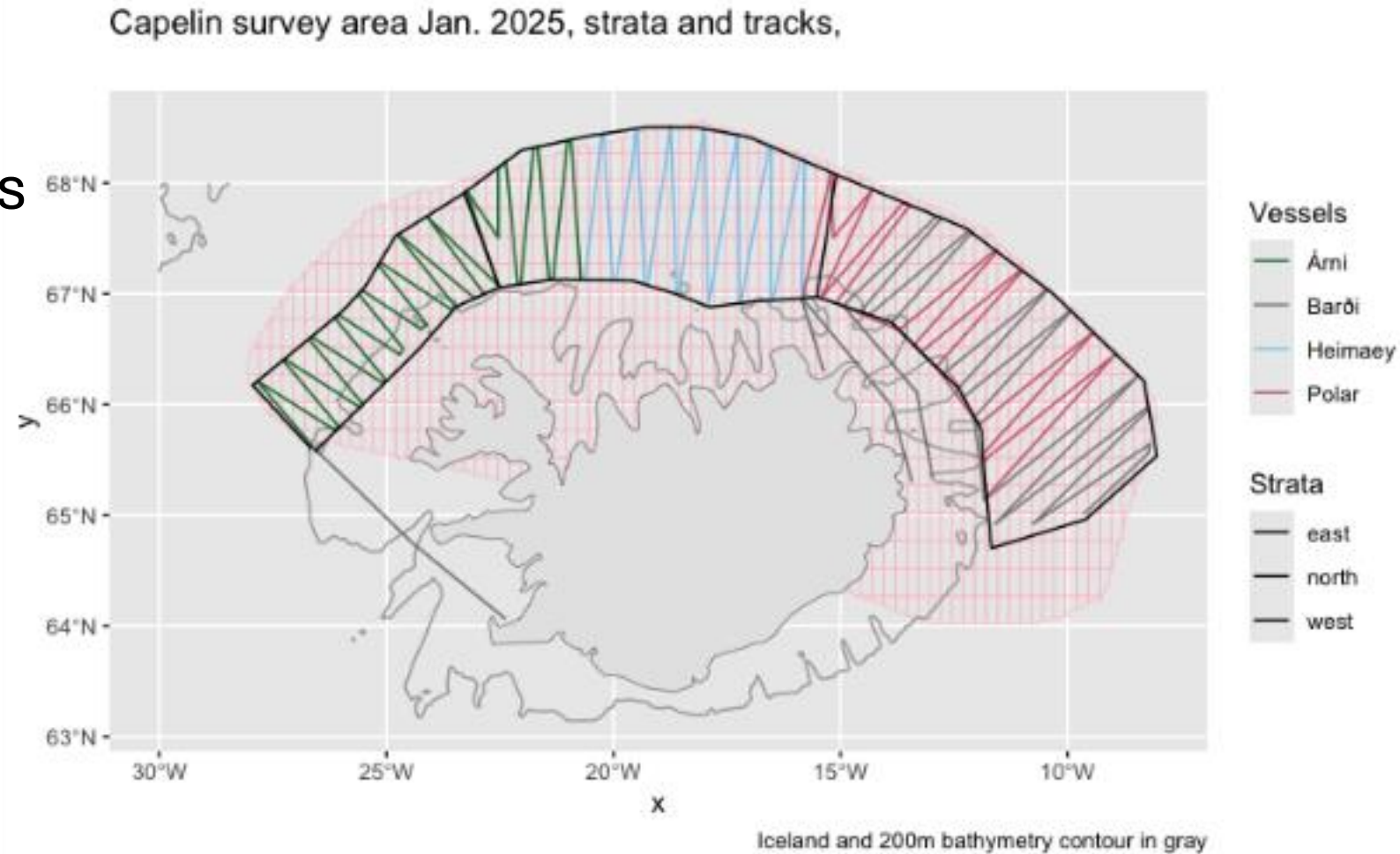
Boundaries:

- Extending ~5 nmi further than capelin dist
- 200 m isobath on the shelf

Months Jan/ Feb

Survey design main Jan coverage

- Predetermined transects
- Sea ice coverage and prediction taken into account in **NV**
- Split into **3 strata**
 - Northwest – **green**
 - North – **blue**
 - Northeast and east – **darkred and grey**



Survey design

February

● Feb 8 to 19

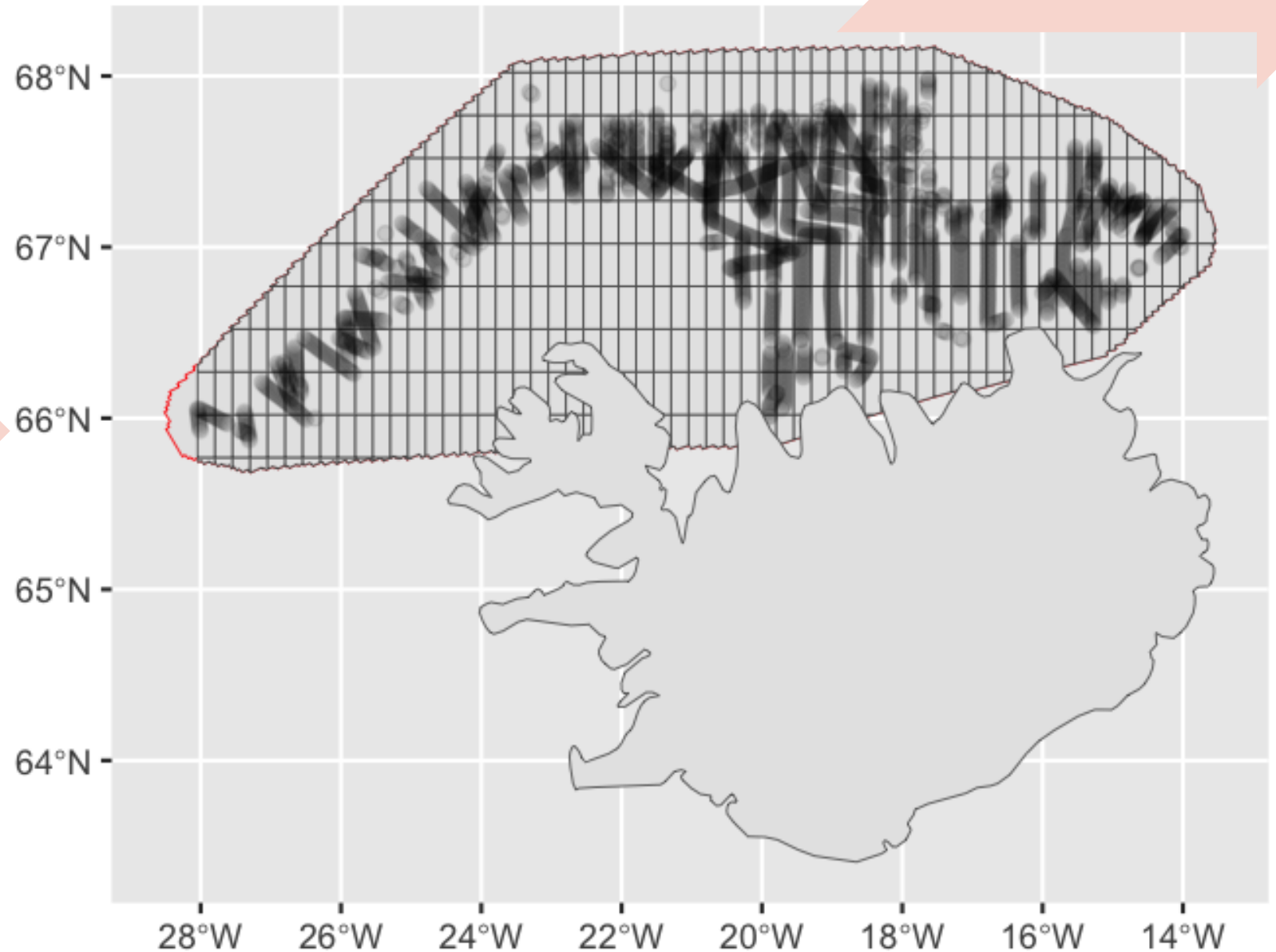
Analyse old data (2000-2021)

Select the survey grid area based on the occurrence of capelin acoustic backscatter ($nasc > 0$).

Limits:

- *5 nmi outside of that boundary

Month of February



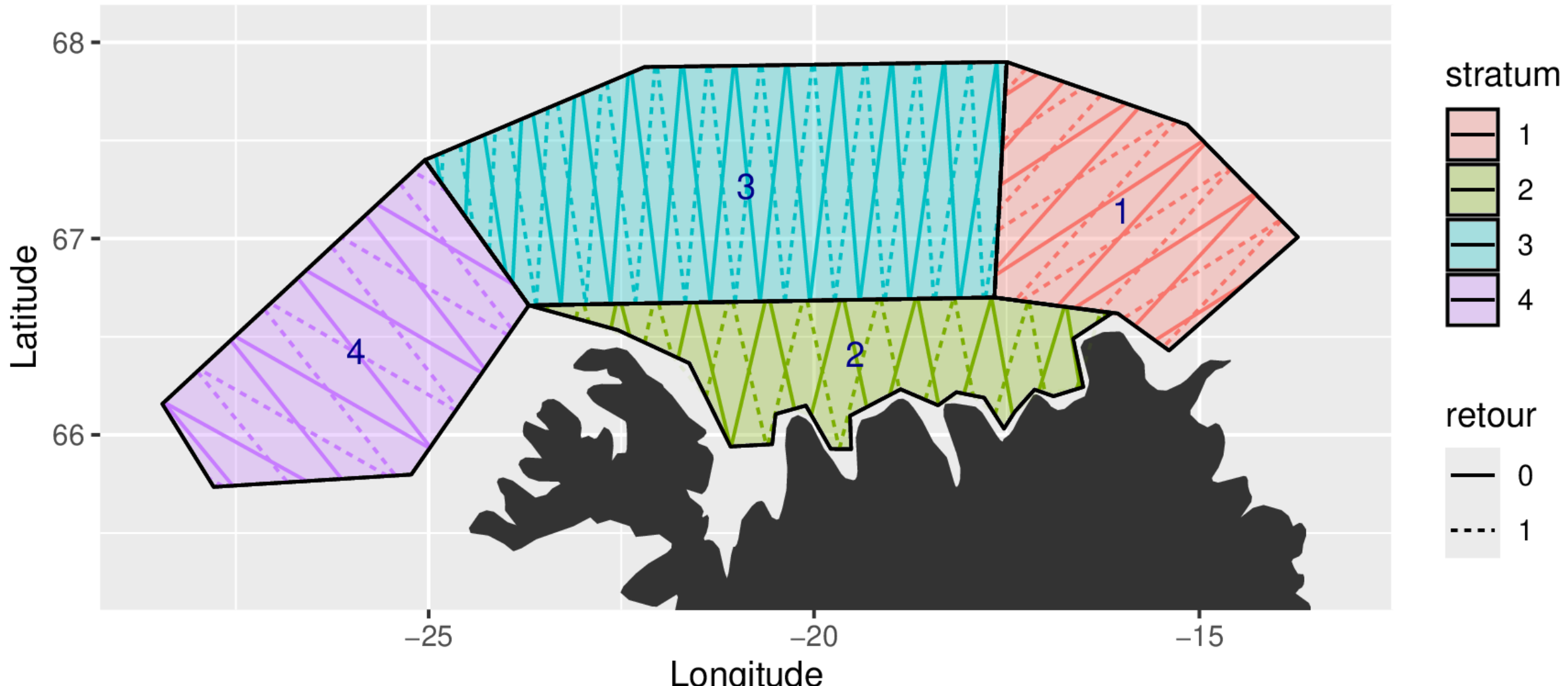
Offshore: high-priority
Inshore: Low-priority

Feb 8-19

Survey tracks for different areas

Stratum priority 3, 1, 2 and finally 4

Byggt 'surveyPlanner' í pakkanum <http://github.com/Sea2Data/Rstox>



Survey execution

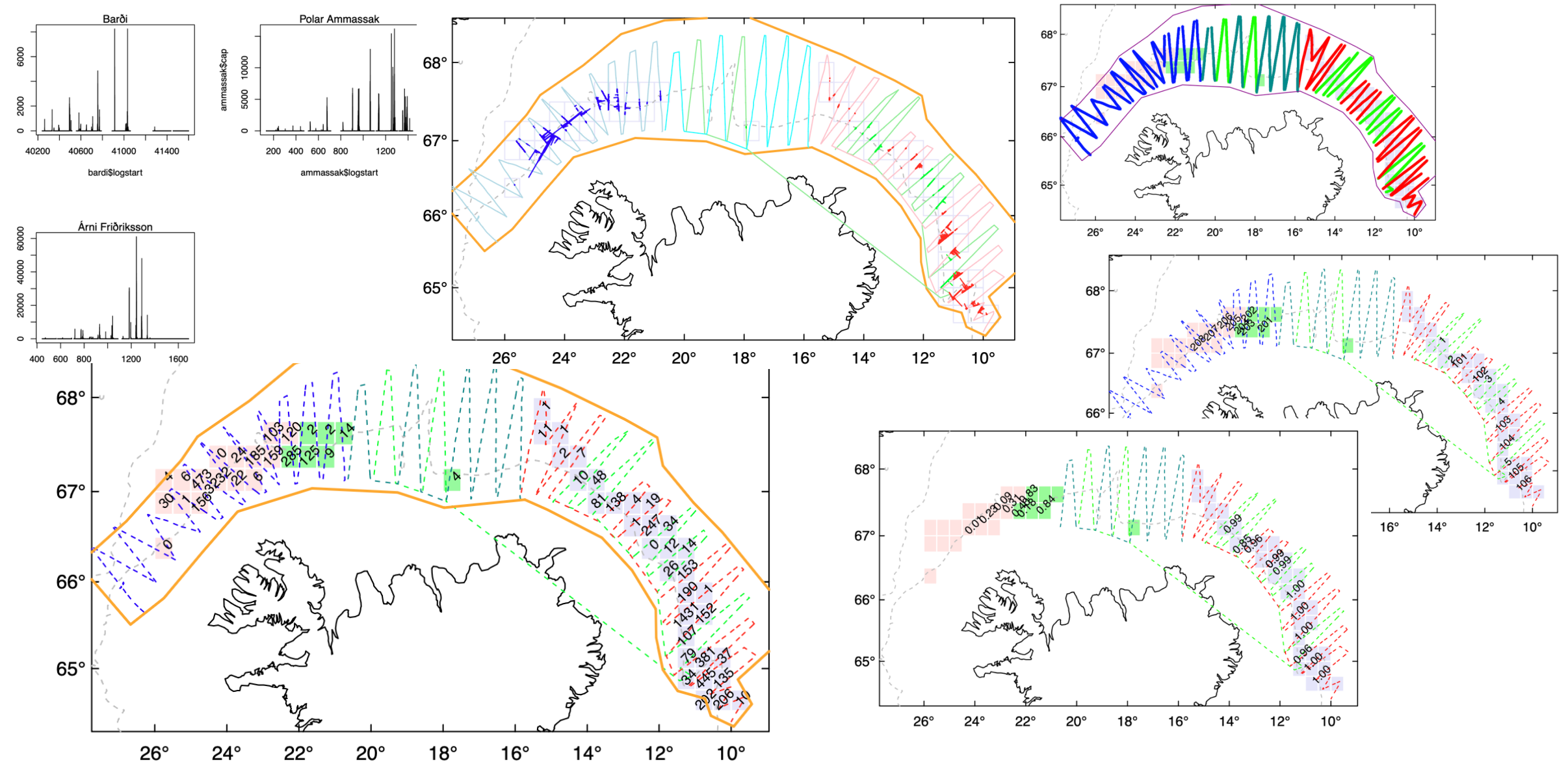
- Highly dependent on weather.
- Depends on number of vessels, not always clear until with ‘short notice’ (especially dependent on whether a quota has been issued).
- Important to plan ahead and try to follow that, while also having flexibility to meet changes in conditions that may arise.
- There are a few examples of adapting density of coverage to distributions observed, we have experimented with different rectangle sizes in the calculations, experience equivocal... .
- As a rule we would have liked more time or vessels and better weather to conduct both winter and autumn IEGJM capelin surveys

Analysis

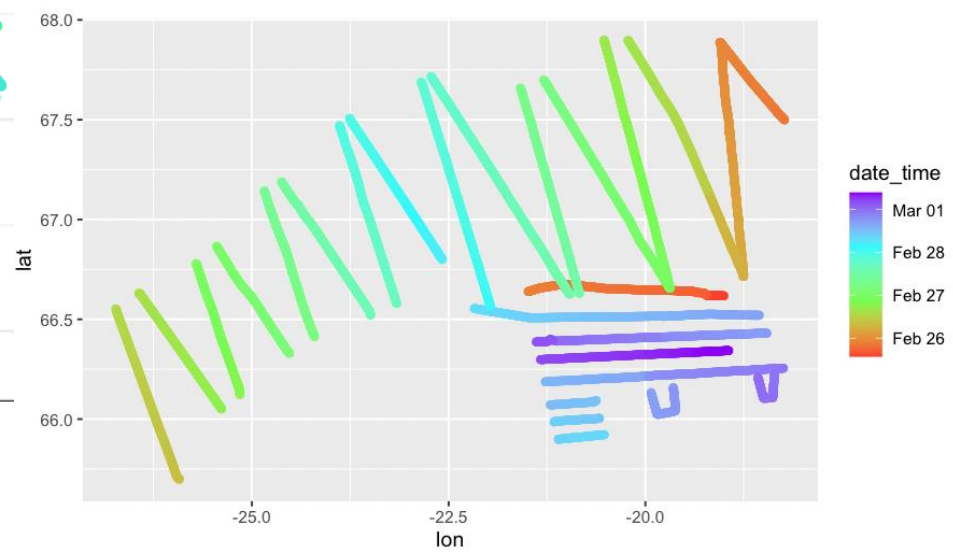
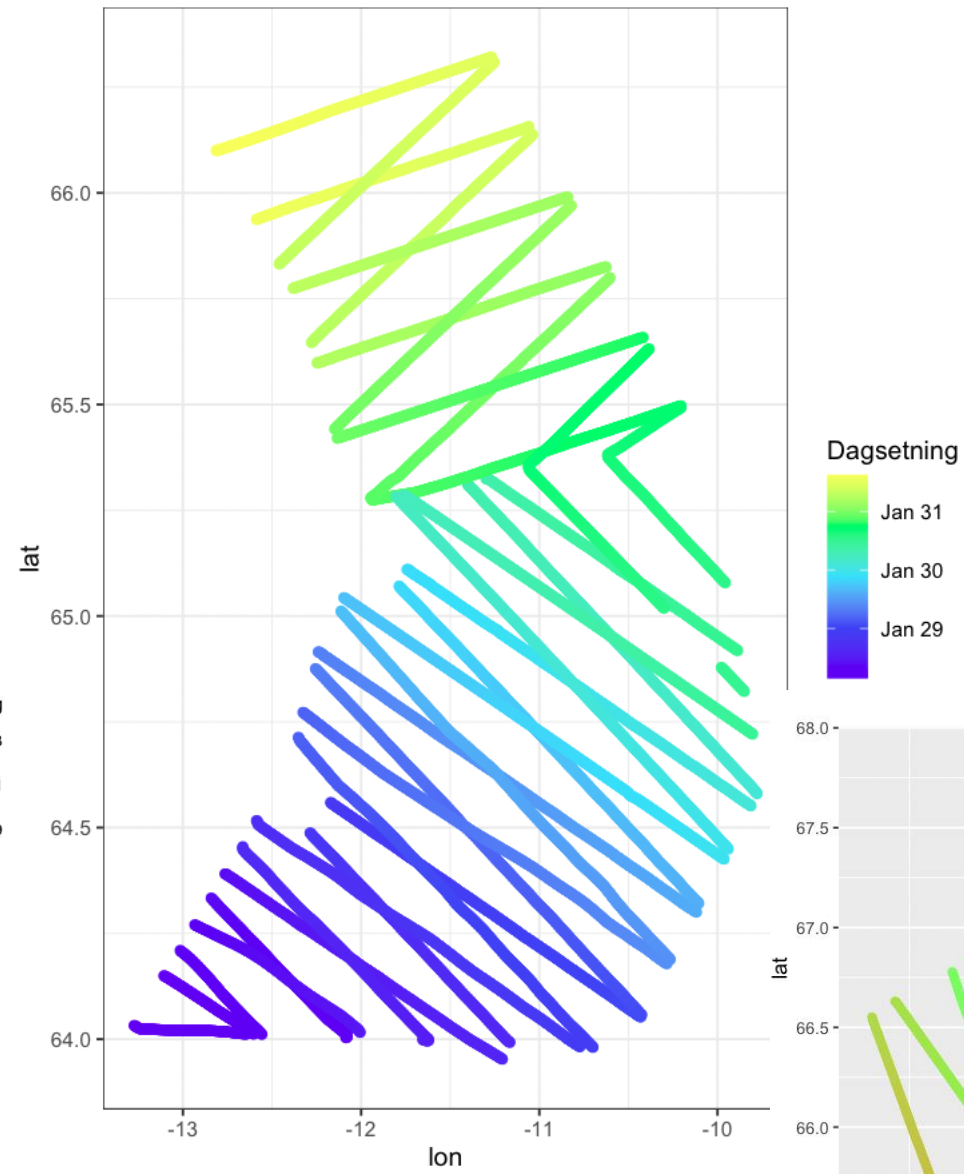
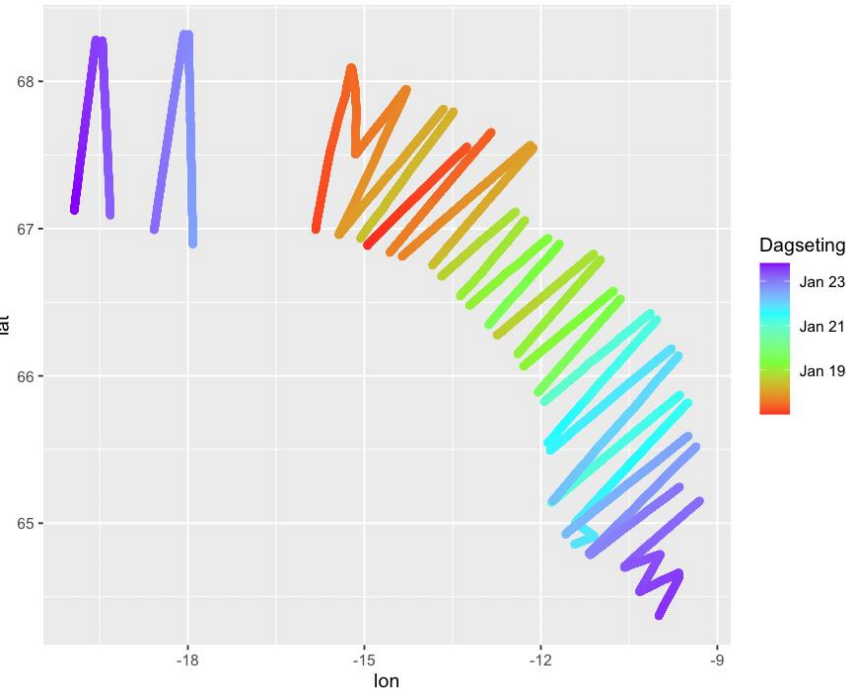


From previously distributed set of figures

EDSU NASC – 0.1 nmi bins



Additional figures



Calculation method according to WKICE (2015) reviewed by WKCAPELIN (2023), ICES benchmark groups

Area density of fish in numbers of fish per square nautical mile is determined from the area back scattering coefficient (in m^2/nm^2) attributed to the species being measured and the average back-scatterer in rectangle i

$$N_{A_i} = S_{A_i} / \overline{\sigma}_i \quad (3)$$

and this is split on length groups and raised to total area of the rectangle with

$$N_{il} = p_{il} N_{A_i} P_{A_i} A_i \quad (4)$$

where P_{A_i} is proportion of square i 'guesstimated' to 'contain' fish and A_i its area.

All samples used in stratum j are used with equal weight to set up an age length key and average weights and maturity by length yielding numbers

$$N_{ial} = N_{il} n_{jal} / \sum_l n_{jl} \quad (5)$$

biomass

$$B_{ial} = w_{jl} N_{ial} \quad (6)$$

and SSB

$$SSB_{ial} = m_{jl} B_{ial} \quad (7)$$

by age and length in square i in area j .

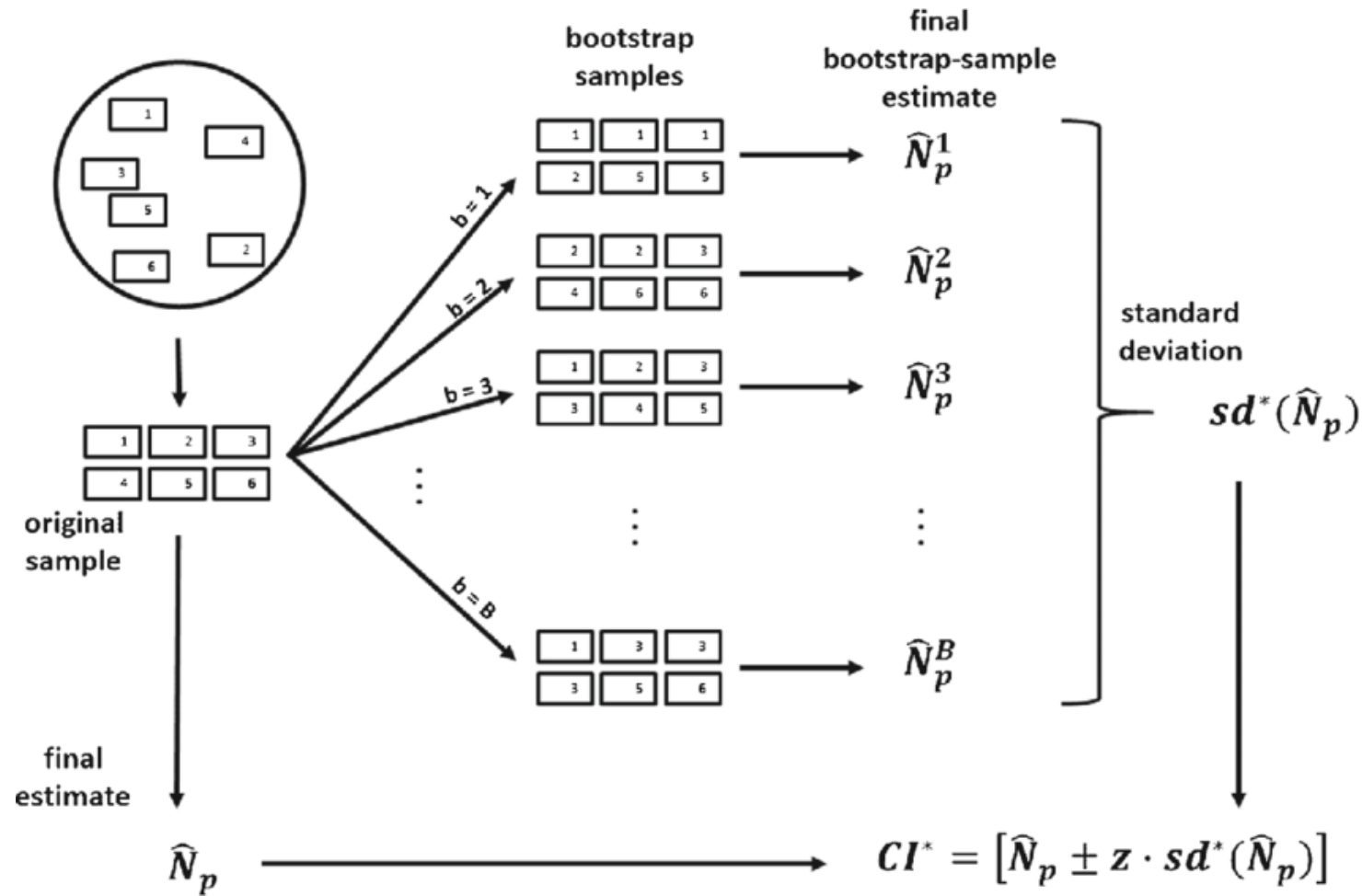
Bootstrap method used to evaluate uncertainty

- For each stratum included in estimation we sample with replacement
 - rectangle echo abundance (EA),
 - trawl stations and
 - individuals on stations

giving same number of 1) rectangle EAs, 2) stations and 3) fish on selected stations as in observed on survey (for one bootstrap replicate)

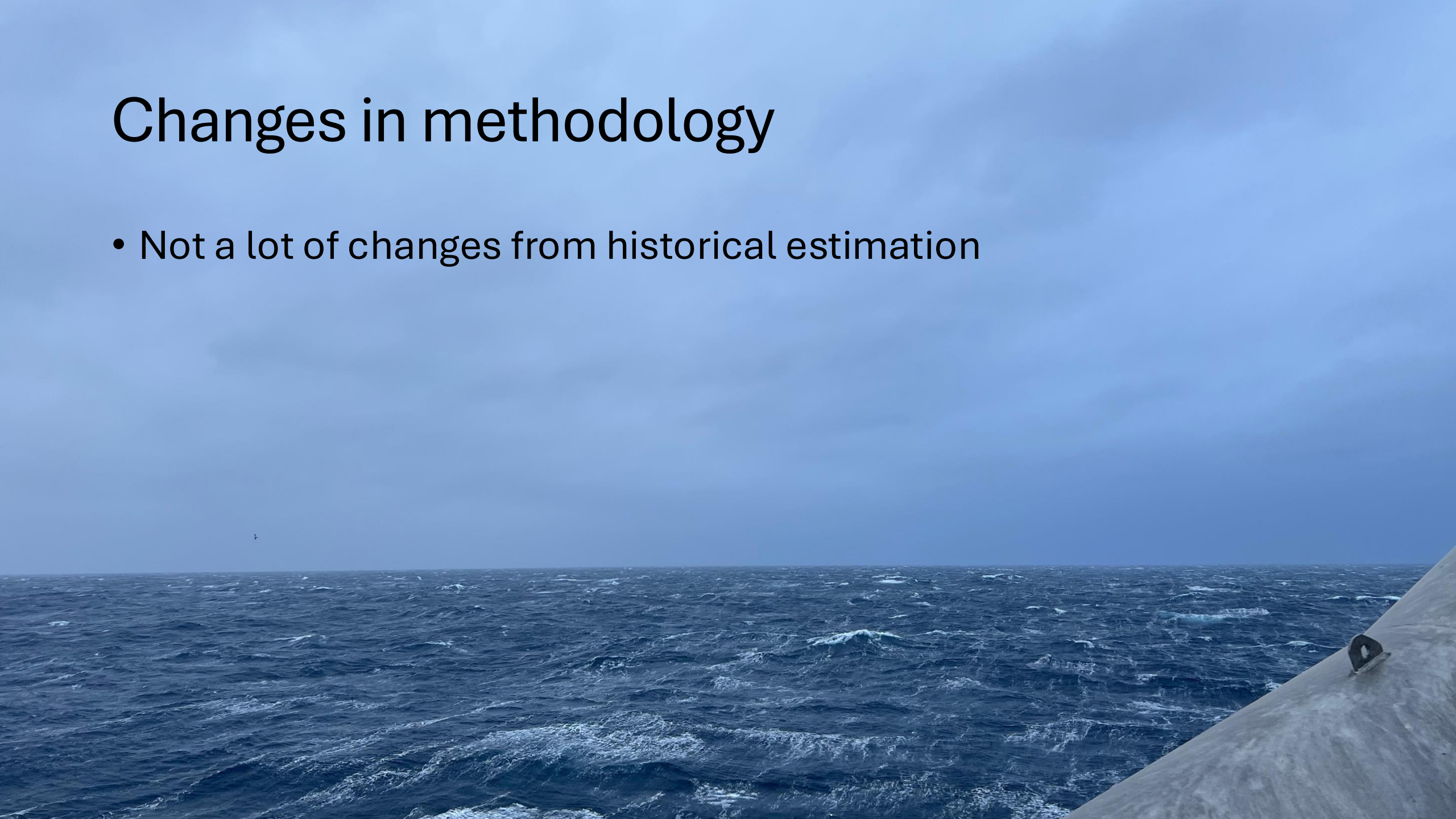
- Rectangle EAs, trawl stations and fish can occur repeatedly
- We estimate from every replicate (increasing to 100 thous. bootstrap replicates, from initial 10 thous.)
- Sum of stratum bootstraps gives total estimate
- Gives bootstrap distributions or uncertainty for selected (any) stock parameter, for the total estimate and by area

Bootstrap schematic



Changes in methodology

- Not a lot of changes from historical estimation



'Changed methodology'

- HV (grand old man of capelin) estimated numbers by rectangles but age- and maturity composition and biomass by area ('nordat' program), after 2010 numbers, age, maturity and biomass been the same for all rectangles in a stratum.
- Sample allocation was not always the same, sometimes *ad hoc*.
- Improved post-processing programs, resolution in echo integration increased, from typical 1 nmi bin to 0.1 nmi, occurred at approx. same time.
- Changed procedure on edges of coverage, now applying a perimeter (outline of coverage used as boundary), previously when rectangle were only partly covered area allocated to capelin distribution was reduced by approx. proportion (as $\frac{1}{2}$, $\frac{1}{4}$, etc).
- Surveys have changed from mainly conducted on 1 or 2 r/v-s, to collaborative surveys with increased number of vessels, but executed more quickly.
- Number of coverages in a season has increased after collaborative surveys became more frequent.

Conclusions

- More rigour in planning and execution may be needed
- Protocol for dealing with differences in stratum coverage
 - have done a few estimates with mixed rectangle sizes , equivocal experience.
- Strengthening the survey with uncrewed platforms (eg USVs or AUVs)
 - Not a lot of progress has been made, understaffed technically.
- Exploring different estimation methods has been discussed (but not materialized yet):
 - [StoX](#)-approach, similar to ours / ours to theirs, but instead of rectangles transects are used as the object of sampling (BSUs) with replacement in bootstrap for uncertainty.
 - Others have also been mentioned, e.g. ‘geostatistík’.

Thanks

- Thank you for the attention
- Þökkum samstarfsfólki á Hafró, áhöfnum og útgerðum skipa í samstarfsleiðöngrum
 - í vetur og í gegnum tíðina